

Program	B. Tech.	Semester	I
Course	Advanced Engineering Mathematics	Course Code	MATH 1059
Session	July-Dec 2025	Unit I	Differential Calculus

1. Verify Rolle's theorem for the function $f(x) = x^2$ in the interval $[-1, 1]$.
2. If c is a real constant, prove that the equation $x^3 - 6x^2 + c = 0$ cannot have distinct roots in $[0, 4]$.
3. If $f(x), g(x), h(x)$ are differentiable in (a, b) , prove that there exists at least one point c in (a, b) such that

$$\begin{vmatrix} f(a) & g(a) & h(a) \\ f(b) & g(b) & h(b) \\ f'(c) & g'(c) & h'(c) \end{vmatrix} = 0.$$

4. Determine c of Lagrange's mean value theorem for the function $f(x) = x + \frac{1}{x}$ in $\left[\frac{1}{2}, 3\right]$.
5. Using Lagrange's Mean Value theorem, prove that $\frac{\cos a\theta - \cos b\theta}{\theta} \leq (b - a)$, if $\theta \neq 0$.
6. Determine c of Cauchy's mean value theorem for the following pair of functions in $[a, b]$:
 $f(x) = e^x, g(x) = e^{-x}$.
7. If $f(x) = \frac{1}{x^2}$ and $g(x) = \frac{1}{x}$, prove that c of Cauchy's Mean Value theorem is the harmonic mean between a and $b, a > 0, b > 0$.
8. Find the n^{th} derivative of following functions:
 (i) $\sin 2x \cos 3x$ (ii) $e^x \sin^2 x$ (iii) $\log_e(ax + x^2)$ where $(0 < a < x)$.
9. If $y(x) = a \cos(\log_e x) + b \sin(\log_e x), (x > 0)$ then show that
 $x^2 y_{n+2}(x) + (2n + 1)xy_{n+1}(x) + (n^2 + 1)y_n(x) = 0$.
10. If $y(x) = e^{m \sin^{-1} x}$, then prove that
 $(1 - x^2)y_{n+2}(x) - (2n + 1)xy_{n+1}(x) - (n^2 + m^2)y_n(x) = 0$ and hence find $(y_n)_0$ i.e. $y_n(x)$ evaluated at $x = 0$.
11. Expand $f(x) = x^5 - x^4 + x^3 - x^2 + x - 1$ in powers of $(x - 1)$. Hence find $f(0.99)$.
12. Expand $\tan\left(x + \frac{\pi}{4}\right)$ as far as the term x^4 and hence evaluate $\tan 46.5^\circ$.
13. If $u(x, y, z) = (x^2 + y^2 + z^2)^{-1}$, then find the value of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$.
14. If $u = y^2 e^{y/x} + x^2 \tan^{-1} \frac{x}{y}$, then show that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u \text{ and } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2u.$$

15. If $u = \frac{1}{3} \log_e \left(\frac{x^3 + y^3}{x^2 + y^2} \right)$, where $x > 0, y > 0$ then prove that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{3} \text{ and } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -\frac{1}{3}.$$

16. If $x + y + z = u, y + z = uv, z = uvw$, show that $\left| \frac{\partial(x,y,z)}{\partial(u,v,w)} \right| = u^2 v$.

17. If $x = uv, y = \frac{u+v}{u-v}$, find $\left| \frac{\partial(u,v)}{\partial(x,y)} \right|$.

18. If $u = x^2 - y^2, v = 2xy$ where $x = r \cos \theta, y = r \sin \theta$, show that $\left| \frac{\partial(u,v)}{\partial(r,\theta)} \right| = 4r^3$.

19. Discuss the points of maxima, minima and saddle points (if any) for the function

$$f(x, y) = x^2 + y^2 + 6x + 12.$$

20. Find the extreme values attained by the function

$$f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x.$$